

Towards High-resolution Simulation of Post-Disaster Economies Utilizing Firm-level Data

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Abstract

Major disasters affect economies in highly complex ways due to the strong interrelationship of economic entities to each other and their heavy dependence on infrastructures. As an example, damages to the lifeline and transportation networks, accessed by firms and their suppliers and consumers, brought upon by disasters are highly heterogeneous, causing extremely varied impacts on different firms even if they are located within the same geographical region. Moreover, the interdependence of economic entities makes these disaster impacts ripple through the regional and national economies in another highly complex manner. For this reason, accurate assessment of the short-term and long-term economic impacts of disasters and the economic performance of their corresponding candidate recovery plans calls for an approach that considers individual economic entities, their complex interdependence, and other real-world constraints such as access to lifelines, transportation, and changes in the government policies during the post-disaster recovery period. Achieving this level of analysis requires high-resolution modeling of the economy and the infrastructure as an integrated system. Going towards this objective, we developed a high-performance computing (HPC) extension for an agent-based economic model that is capable of efficiently simulating large economies with hundreds of millions of economic entities at a 1:1 scale. In our most recent works, we further advanced this model by integrating it with the historical business data of around 1.4 million firms provided by Teikoku Databank, Ltd., such that the firm agents in the model are initialized with the data of their real-world counterparts. Validation tests show that the model is capable of reproducing the past observed indices of the Japanese economy at three levels of resolution: national (e.g., national GDP), industrial (e.g., sectoral production) and firm-level (e.g., sales records of major firms). To demonstrate the application of the developed system in a disaster case and policy use, we estimated the economic impacts of a hypothetical disaster scenario and simulated a simple recovery plan. This research contributes to the understanding of disaster consequences and highlights the use of firm-level data in the field of disaster risk analysis and management.

Keywords: Agent-based economic models, Post-disaster economic simulation, High-Performance Computing

1. INTRODUCTION

The strong interrelationship of economic entities, such as firms and households, and their dependence on infrastructures make the impacts of major disasters propagate through the economy of the entire country, causing secondary economic losses, even if the disaster is geographically localized. As an example, while the Nankai-trough Earthquake is estimated to cause 170 trillion JPY in direct damages such as damages to buildings and infrastructure over a localized region, the estimated economic losses over a 20-year period exceeds 1.2 quadrillion JPY [1]. To enhance disaster resilience, it is crucial to be able to comprehensively assess the long-term economic impacts of disasters and their corresponding candidate recovery plans considering the complex interdependencies of the economic entities.

Studying the economic impacts of disasters has been an area of interest over the past few decades. Disasters create highly heterogeneous conditions on different economic entities, even if those entities belong to the same industry or are located in the same region. However, the traditional approaches to estimating the economic impacts of disasters, such as computable general equilibrium models and input-output models, are highly coarse-grained as they lump a large number of economic entities with common characteristics into a single representative entity, rendering these models incapable of considering the highly heterogeneous conditions experienced by different economic entities. To accurately capture the economic impacts of disasters considering these conditions, each individual economic entity and their interactions with each other must be considered in the economic model at fine resolution.

Agent-based economic models (ABEMs) address the above-mentioned limitations of traditional models by simulating economies at a fine resolution as a time-step and event-driven system of interacting heterogeneous agents. Several studies have used agent-based models to study various economic problems, such as assessing the effects of capital and credit available to consumer goods producing firms [2], investigating the impacts of credit networks on growth rate [3], and estimating the economic damages caused by a flood disaster [4]. While the ABEM developed by Poledna et al. [4] [5] is already capable of simulating economies at a 1:1 scale, it is not computationally capable to simulate large-scale economies with hundreds of millions of agents. Diessenberg et al. [6] attempted to use parallel computing to simulate the Eurozone economy with hundreds of millions of agents, but their implementation only considered a small hypothetical labor market.

Realizing the need for a High-Performance Computing (HPC) enhanced ABEM to simulate large economies with fine-grained details and conduct comprehensive assessments of post-disaster economies, we developed an HPC extension for ABEMs that is capable of efficiently simulating large-scale economies with hundreds of millions of agents at 1:1 scale. While the developed HPC-enhanced framework can be applied to any ABEM, the model developed by Poledna et al. [5], which has been validated by comparing model forecasts with past observed data, was used as the base model. Furthermore, we integrated this model with the historical business data of around 1.4 million firms provided by Teikoku Databank, Ltd. (TDB), such that the firm agents in the model are initialized with the data of their real-world counterparts.

In this short paper, we present some details of our HPC extension and the application of the new implementation to simulate the Japanese economy. The validation of the model at the national and sectoral levels were done by comparing the simulation results with the observed national macroeconomic indices and sectoral production in the Japanese economy. The model was also validated at the firm-level by comparing the simulation outputs with the historical sales records of major firms. Finally, we demonstrate an application of the developed framework using hypothetical disaster case scenarios, highlighting the potential use of the simulator in studying the economic impacts of disasters.

2. HPC EXTENSION OF THE AGENT-BASED ECONOMIC MODEL

2.1. Overview of the ABEM

An agent-based economic model is a time-step and event-driven system of interacting autonomous agents that try to mimic the activities and interactions of their real-world counterparts. The agents in the model represent various economic entities, such as individual households, firms, banks, and

government. Each agent's behavior and its interaction with other agents are defined by a set of rules that are mathematically well-defined based on behavioral economics. The agents interact with each other, making decisions and acting based on the information available to them. The ABEM is initialized only at the base period, after which the economy evolves depending on the millions of complex interactions of its agents.

A schematic diagram of a typical ABEM consisting of various types of agents is shown in Figure 1. The types of agents include firms, investors, active and inactive households, banks, the central government, and the central bank. The model mimics various real-world economic interactions such as buying and selling (i.e., goods market) by allowing the agents to interact with each other. The firms hire workers based on their labor requirements and pay wages to these workers. The firms also buy capital and intermediate goods from firms of other industries depending on their production requirements. Foreign sellers import goods from external economies and foreign buyers export products to external economies. The firms then produce goods considering their production demands and available labor, capital, and raw materials. Aside from active households (i.e., workers), other household types include investors and inactive households. Investor agents represent owners of firms and receive dividends as their income. Inactive household agents represent the non-working population, such as children, retirees, and persons with disabilities, who receive social benefits from the government. The households buy goods and services from the industry for consumption and investment. The bank keeps deposits from firms and households and provides loans to them when requested. The government consumes various goods and services from the industry, provides social benefits, and collects taxes, while the central bank manages the monetary policies and policy rates. Millions of these processes happen at each time step of the simulation, making the economy evolve over time depending on the interactions and decision-making of the participating agents.

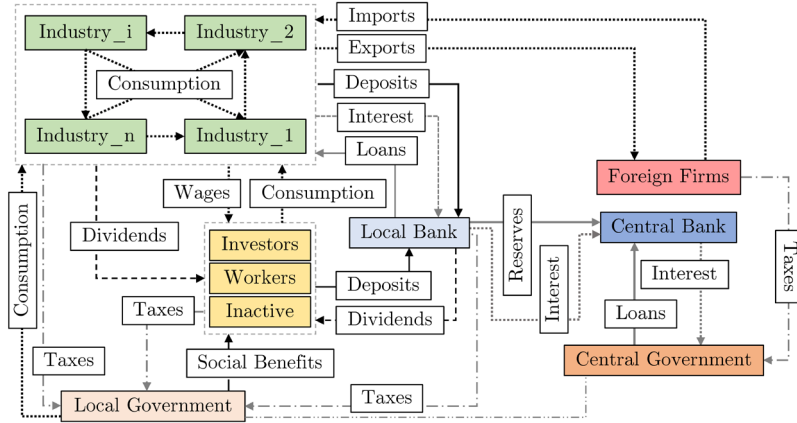


Figure 1. Schematic diagram of a typical ABEM

2.2. Scalable HPC Extension

The major challenge in developing a scalable HPC implementation is the complicated interactions of the agents on centralized graphs such as the credit network and scale-free randomly growing graphs such as the labor market and the goods market. We addressed this problem by partitioning the agents based on a representative graph of the labor market and by employing various computational strategies to minimize the number of MPI communications. Furthermore, we eliminated performance bottlenecks caused by memory-intensive operations by implementing cache-friendly data structures and algorithms. We conducted scalability test simulations with 330 million agents and our tests showed that the implementation has 70% strong scalability on 128 CPUs using the Oakbridge-CX supercomputer at the University of Tokyo. The short execution time of 110 seconds per one period shows that our HPC implementation requires relatively low computational resources. The attained high computational performance makes it possible to conduct Monte-Carlo simulations of large economies at a 1:1 scale, which is essential to address various uncertainties involved within the economic model. Readers interested in the details of the implementation of the distributed memory parallel HPC-enhanced code are referred to the study by Gill et al. [7]

3. DATA GENERATION FOR SIMULATING JAPANESE ECONOMY

The potential of the model to produce reasonably accurate forecasts for an economy can only be realized when the model parameters and the agents' data are derived from real data. The ABEM requires two types of data: (1) parameters that are shared by all agents of a particular class (e.g., industry); and (2) agent-specific data which is unique for each agent. A detailed list of required input parameters for the agent-based economic model is available in the study by Poledna et al. [5] and it is not reproduced here to keep the conciseness of the paper. This section only provides a brief outline on how these data were generated from analyzing data from various sources.

3.1. Data Sources

Since the ABEM is a micro-founded model, the parameters of each agent should be ideally set according to the balance sheet and decision-making behavior of its real-world counterpart (e.g., individual firm). However, data at such fine resolution is either hardly available or too expensive to procure. Although some major companies publish their financial data on a regular basis, no information is available regarding their production decisions. A similar problem is present with the households' data and banks' data.

The next best alternative to the individual economic entity data would be macroeconomic data published by the Japanese government and policy think tanks, from which a consistent set of the required parameters could be extracted and generated. Publicly available data from the following Japanese organizations are the main sources for this study:

- The Cabinet Office of Japan (CAO)
 - <https://www.cao.go.jp/index-e.html>
- Portal site of the official statistics of Japan (e-Stat)
 - <https://www.e-stat.go.jp/en>
- The Bank of Japan
 - <https://www.boj.or.jp/en/index.htm/>
- the Research Institute of Economy, Trade, and Industry (RIETI)
 - <https://www.rieti.go.jp/en/>

The 2015 Input-Output (IO) Table [8] at producers' price for Japan, which can easily be accessed from e-Stat, is the major source of input data for simulating the Japanese economy with the ABEM. The IO Table describe the demand and supply relationships between the producing and consuming entities within the economy. The parameters defining the interdependencies between the industries and the consumption behavior of households, government, foreign buyers, and foreign sellers are also extracted from the IO Table. Table 1 shows a representative IO Table and the model parameters obtained from the IO Table.

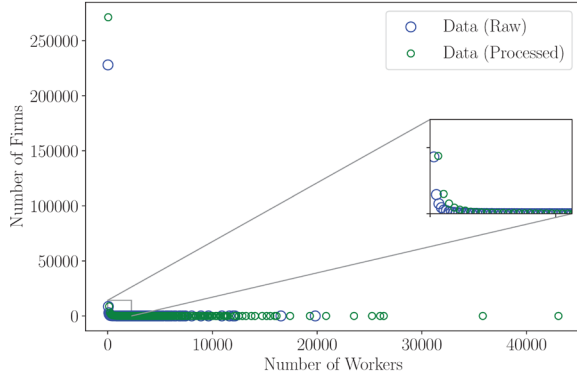
In generating the industry-specific parameters, the System of National Accounts (SNA) from CAO and the Sector-wise Employment data from e-Stat are also used. The SNA contains useful information such as the total capital, bank deposits, and bank loans of the firms for each industry. On the other hand, the number of workers in each industry is obtained from the Sector-wise Employment Data. The Japanese IO Table [8] was set to be the base data for the study and all other datasets were normalized to the IO Table data to maintain consistency among different datasets.

In our most recent endeavors, Teikoku Databank, Ltd. provided us with the historical sales and labor records of about 1.4 million firms, which contain data from 2000 to 2018. This was used to initialize the parameters of the firm agents to match their real-world counterparts. While the data provided by TDB offers information for the majority of Japanese firms, it has several inconsistencies with the macroeconomic data. For example, the data uses a different sector classification and has a different sector-wise total number of firms and workers compared to the data obtained from e-Stat. To ensure seamless integration of the firm-level data with the agent-based simulator and other input data, multiple preprocessing steps, such as sector matching and data cleaning, were applied to the data while also intentionally preserving the nature of the original data. The distribution of the firm sizes by labor was used to determine if the characteristics of the raw data were maintained. Figure 2 shows the firms'

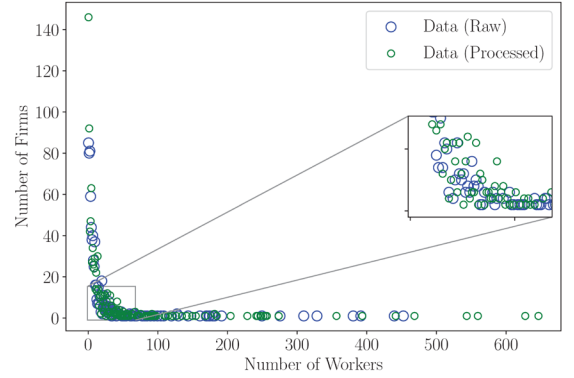
labor size distribution before and after processing the data for various sizes of sectors. It can be observed from the plots that the processed data closely resembles the distribution of the original firm-level data.

Table 1. Overview of the IO-table

		Supply sector (seller)					Intermediate demand					Final demand											
		Sector-1	Sector-2	...	Sector-n	Total	Consumption expenditure outside households	Private consumption expenditure	Consumption expenditure of general government	Consumption expenditure of general government (social fixed capital)	Gross domestic fixed capital formation (public sector)	Gross domestic fixed capital formation (private sector)	Increase in stocks	Exports	Total	(less) Imports	Domestic Production						
Demand sector (buyer)	Intermediate input	Sector-1	m_{11}	m_{21}	...	m_{n1}	A_1	h_1^1	h_1^2	g_1	c_1^1	c_1^2	c_1^3	s_1	e_1	d_1	i_1	y_1					
		...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...					
		Sector-n	m_{1n}	m_{2n}	...	m_{nn}	A_n	h_n^1	h_n^2	g_n	c_n^1	c_n^2	c_n^3	s_n	e_n	d_n	i_n	y_n					
		Total	M_1	M_2	...	M_n	A	H^1	H^2	G	C^1	C^2	C^3	S	E	D	I	Y					
Gross value added	Consumption expenditure outside households	w_1^1	w_2^1	...	w_n^1	W^1																	
	Consumption of employees	w_1^2	w_2^2	...	w_n^2	W^2																	
	Operating surplus	π_1	π_2	...	π_n	Π																	
	Consumption of fixed capital	k_1^1	k_2^1	...	k_n^1	K^1																	
	Consumption of fixed capital (social fixed capital)	k_1^2	k_2^2	...	k_n^2	K^2																	
	Indirect taxes (except custom duties and commodity taxes on imported goods)	t_1^Y	t_2^Y	...	t_n^Y	T^Y																	
	(less) Current subsidies	t_1^P	t_2^P	...	t_n^P	T^P																	
	Total of Gross value added	GVA_1	GVA_2	...	GVA_n	GVA																	
Total Domestic Production		y_1	y_2	...	y_n	Y																	



(a) Commerce Sector



(b) Leather products sector

Figure 2. Firms' labor size distribution before and after data processing for varying sector sizes

3.2. Parameters of Individual Firm Agents

The initialization parameters required by the firm agents include number of employees, production, productivity, wage, capital goods, input goods, and inventories. The initial number of employees and production of each individual firm were set according to the preprocessed firm-level data, and the rest of the firm-specific parameters were derived based on these two parameters. Since the initialization parameters of the firm agents are now unique to each individual firm agent, the corresponding decision-making rules of the agents, such as the rules for deciding the quantity of production for the next period, are also set to match the characteristics of the agents' real-world counterparts.

There are some industrial sectors, such as the tobacco sector and the public administration sector, where the utilization of firm-level data is not applicable due to various reasons. Some reasons include unsuitable data (e.g., having an unrealistic labor size distribution in that sector) and mismatched sales definitions (e.g., the TDB's definition of recorded sales for that specific sector is different from that of e-Stat). For these cases, we generated the firm agents' parameters by disaggregating the data obtained from the publicly available macroeconomic data. Since the quantity of goods produced by a firm is directly related to the number of workers the firm employs, the size distribution of firms can be obtained from the labor size distribution among firms. The e-Stat provides the sector-wise labor size distribution of Japanese firms for 2016. The latest available IO Table for Japan is the 2015 table. Therefore, the base year of the dataset for this research is set to 2015. However, the labor size distribution of firms for 2015 is not available, and the closest available is the one for 2016. Since firm size distribution does not change much within a year, it was assumed that the firm size distribution in 2015 is the same as that in 2016. The industrial classification used in the enterprise census data for 2016 is also different from that in the 2015 IO Table. To resolve this mismatch, the firms and labor distribution for the sectors of the 2015 IO Table were derived by merging or splitting relevant sectors of the 2016 enterprise census data according to the definition of each sector in both datasets.

3.2. Parameters of Government

The government agent requires specific parameters, including the consumption budget, various tax and subsidy rates, subsidies, and unemployment benefits paid to households, as well as the tax rate set by the central bank. The process of generating data can be outlined as follows.

- (1) The government distributes its budget for spending across different sectors through its agencies. While the exact number of agencies is not explicitly available from the data, it is assumed to be 25% of the total number of firms in the economy.
- (2) The government's consumption budget is treated as an exogenous parameter and determined based on the SNA data.
- (3) Indirect taxes and subsidies for industries are derived from the IO table. The consumption tax rate and social benefits received by households are extracted from household income, expenditure, and saving data along with IO table data. Missing rates for taxes such as export tax and corporate tax are assumed to be zero.
- (4) Government debt (L^G) is obtained from the SNA data. The interest rate on loans (r^G) is assumed to be zero due to a lack of available data.

The required parameters for the remaining agents: households, foreign firms, and the central bank are also obtained from the above-mentioned sources. The data generation process for these agents is not presented here to keep the brevity of the paper.

4. VERIFICATION AND VALIDATION OF THE AGENT-BASED SIMULATOR

The process of verification and validation is essential for ensuring the accuracy, reliability, and credibility of simulation models. Verification involves comparing the simulation outputs with the results obtained from a previously verified implementation, ensuring that the new implementation is free from errors and reproduces the expected results accurately. On the other hand, validation aims to assess the model's ability to replicate real-world behavior by comparing the simulation results with past observed data. We verified our HPC-enhanced implementation by comparing the simulation results with the results from Dr. Poledna's implementation [5] in MATLAB.

For the validation of the implemented HPC-enhanced ABEM, we simulated the Japanese economy and compared the model results with the observed Japanese macroeconomic indices and the observed historical sales data of major firms to check whether the new implementation can reproduce the real-world observation with a substantial degree of accuracy. For this, we used the input parameters for the Japanese economy that we prepared as discussed in Section 3. In the problem settings, the model parameters and the initial parameters of the agents are generated according to the process mentioned in the previous section using the Japanese economic data of Q1 2015. Table 2 shows the number of various agents in the generated model. Once the agents were initialized in the first period, no additional external

inputs or instructions were given to the model. The economy and markets were allowed to evolve autonomously based on the complex interactions of the agents. The simulation is conducted for 22 periods starting from the first quarter of 2015 and concluding with the second quarter of 2020. One simulation period represents one financial quarter. The results are then compared with the observed national economic indices obtained from the SNA and sectoral production data obtained from the RIETI. Auto Regression with a lag of 1 (AR(1)) process is used for making forecasts of growth and inflation rates. These forecasts are used by the firm agents to make production and pricing decisions. AR(1) models are constructed based on the historical production data of its real-world counterpart, obtained from the firm-level data provided by TDB. There are various randomness-introducing variables in the model, such as in the decision-making of agents and the forecasting of growth and inflation rates. To account for these uncertainties, Monte-Carlo simulations are conducted with 800 samples.

Table 2. Number of various agents in the ABEM for the Japanese economy

Agent Type	Number of Agents
Industrial Sectors	104
Domestic Firms	1.34 million
Foreign Buyer Firms	667,000
Foreign Seller Firms	104
Investors	1.34 million
Workers	61.6 million
Inactive Households	62.4 million
Bank	1
Government Agencies	334,000
Central Bank	1
Total	127.6 million

Figures 3 and 4 show the comparison of the simulation results using the HPC-enhanced agent-based economic simulator with Japanese macroeconomic indices and sectoral production. The dark and light gray regions in the plots represent the $\pm\sigma$ and $\pm2\sigma$ uncertainty bands, respectively. As the figures show, the simulation results are in good agreement with the observed data. While the model did not capture the sharp increase in GDP in the 8th period and the sharp decrease in household consumption in the 3rd and 4th periods, it was able to reproduce the overall trend of the indices with a substantial degree of accuracy. Regarding the comparison at the sectoral level, it was observed that the model was able to make more accurate forecasts for larger sectors such as the medical sector. In contrast, the model exhibited weaker performance in smaller sectors such as the general-purpose machinery sector. It should also be noted that while some of the observed data points lie outside the uncertainty bands, the difference between the simulation and the observation is small relative to the economic index being simulated.

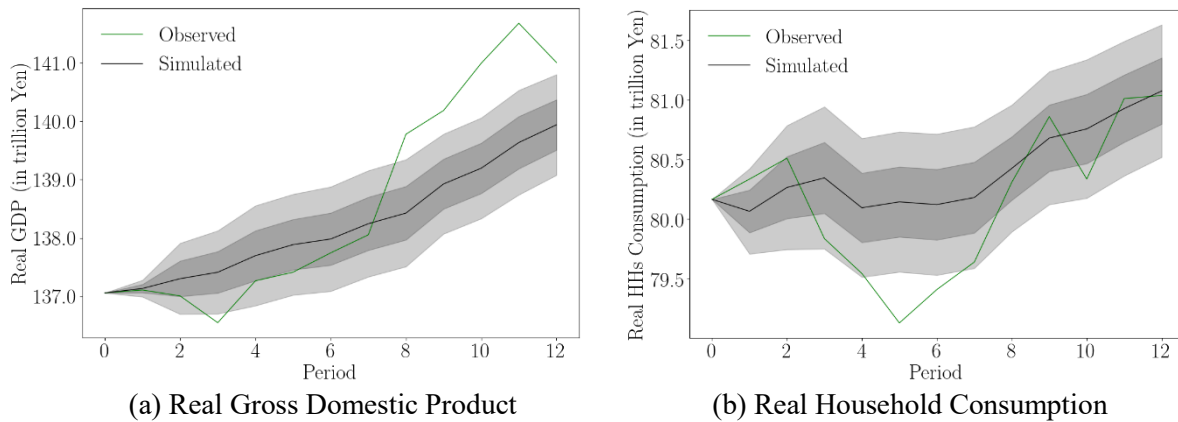


Figure 3. Comparison of model outputs at national level and observed national macroeconomic indices

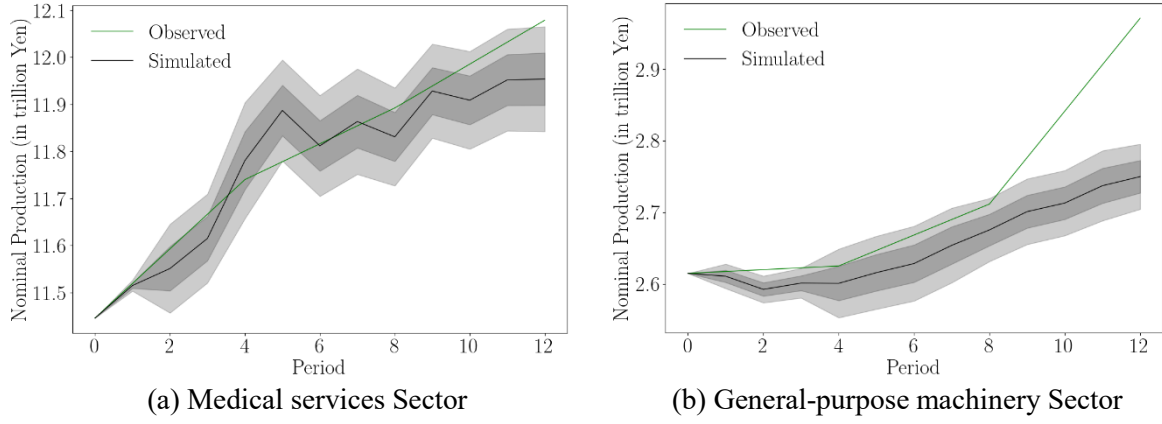


Figure 4. Comparison of model outputs at sectoral level and observed sectoral production data

The integration of the firm-level data with the agent-based economic simulator allows the validation of the model at the firm level, which is the highest level of resolution in this micro-founded approach. To keep the confidentiality of the firm-level data provided by TDB, we randomly selected a firm from each sector's top 5% largest firms based on labor size and compared the simulation results for that firm agent with the historical sales data of its real-world counterpart. Figure 5 shows the firm-level validation for sectors of different sizes. As the plots illustrate, the forecasted production for the randomly selected firms agrees well with the observed data on their real-world counterparts, especially for the firms coming from the larger sectors.

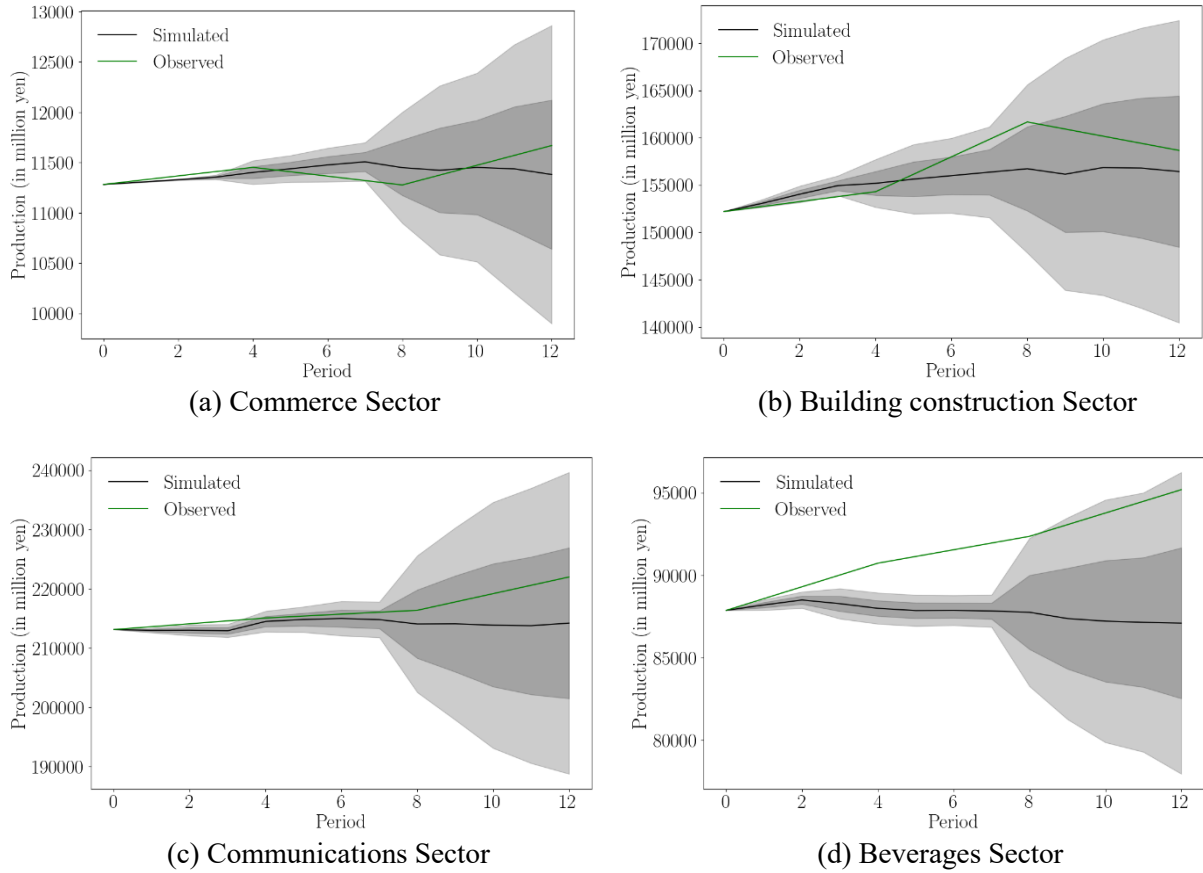


Figure 5. Comparison of model outputs at firm level and observed historical sales data of a randomly selected major firm

To assess the model's performance across all firms within their respective sectors without showing the results for each individual firm, the mean normalized root-mean-square error (NRMSE) was calculated for each sector for each validation period. While there are alternative performance metrics such as mean absolute error or mean absolute percentage error, the root-mean-square error (RMSE) was selected because it emphasizes the effect of larger errors. This choice is more suitable for the ABEM as the simulation errors vary significantly across firms and sectors. Since RMSE is measured in the same units as the target variable (i.e., firm's production), and the firms' production levels differ based on their labor sizes, the RMSE values for each firm were normalized to their firm's respective production values. The mean of the normalized RMSE for all firms within the same sector was then used to evaluate the model's performance for that sector. Equation 1 presents the formula used to calculate the mean NRMSE for sector s and time period t .

$$\overline{NRMSE}_{s,t} = \frac{1}{N_f} \sum_f \left(\frac{1}{Y_{s,f}^{observed}(t)} \sqrt{\frac{\sum_{i=1}^{N_{MC}} (Y_{s,f,i}^{simulated}(t) - Y_{s,f}^{observed})^2}{N_{MC}}} \right) \quad (\text{Equation 1})$$

where the subscripts s , f , and i refer to the index of the sector, the firm of the sector s , and the Monte-Carlo sample, respectively. The total number of samples used in the Monte-Carlo simulation is N_{MC} . The superscripts *simulated* and *observed* refer to the simulated production results and observed production, respectively.

The mean NRMSE and the distribution of NRMSE across firms were plotted for sectors with varying production sizes, as depicted in Figure 6. The figure demonstrates that the mean NRMSE is consistently small, particularly for large-sized sectors, indicating a strong overall performance of the model when considering all firms within each sector. Furthermore, it was observed that the mean NRMSE tends to increase as the period extends further into the future, which is expected as forecasting uncertainties grow larger over longer periods.

Examining the distribution of RMSE, it is evident that the distributions are right-skewed, indicating that more firms exhibit smaller NRMSE values than the mean NRMSE. The plots also include the distribution for firms belonging to the largest 20% of their sectors based on labor size. This distribution reaffirms the previous observations, showing that major firms tend to have smaller NRMSE values and more accurate simulation results. Ultimately, this evaluation shows that the simulator was able to capture the production records at the firm level with a significant degree of accuracy.

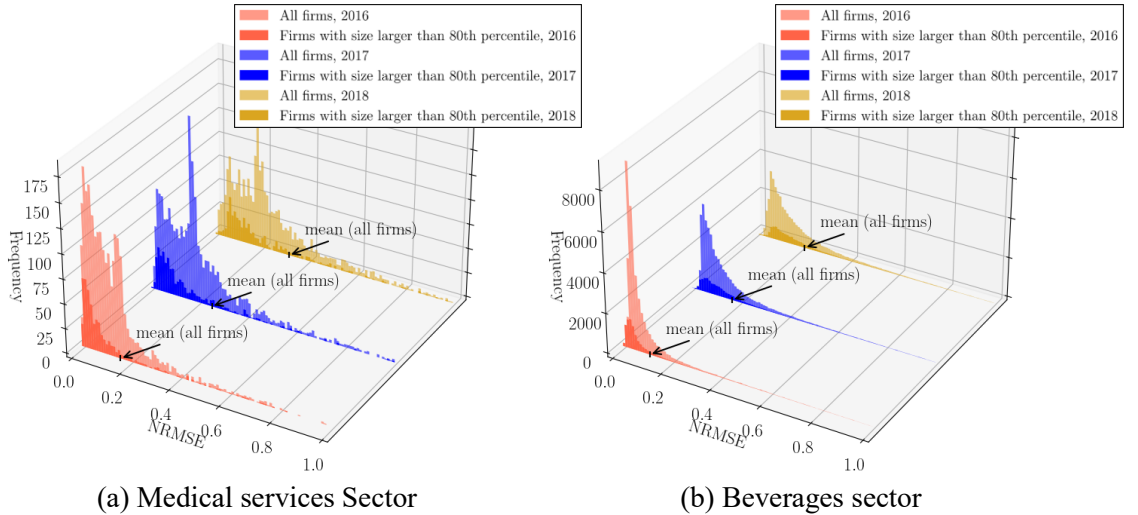


Figure 6. Sector-wise model performance considering all firms

The good agreement of the model results with the past observations indicates that the micro-founded HPC-enhanced ABEM is capable of making reasonably accurate forecasts under ordinary economic conditions.

5. DEMONSTRATIVE APPLICATION WITH HYPOTHETICAL DISASTER SCENARIOS

To demonstrate the application of the developed tool for simulating post-disaster economies, the Japanese economy was simulated under three different scenarios. The first scenario represents normal economic conditions and the other two correspond to post-disaster economic conditions resulting from hypothetical disaster cases. In the disaster scenarios, the capital damages from the 2011 Great East Japan Earthquake (GEJE) were introduced as an economic shock in the first period of the model. The first hypothetical disaster scenario represents a case where the government does not give any financial aid to households, whereas the second scenario depicts a situation where the government provides financial aid to households.

5.1. Agent's Behavioral Rules

The behavioral rules of agents under normal economic conditions are provided in Poledna et al. [5] For the disaster case scenarios, the agents' behavior is assumed as follows:

1. The firms account for the damaged capital as an expense. Thus, their profit falls, impacting their financial stability and their investor's income.
2. The firms try to buy the lost capital starting from the next period after the disaster. However, the availability of loans from the bank and the availability of capital goods in the goods market determine the amount of time it takes for them to recover the lost capital.
3. The households also attempt to recover their lost capital by buying it from the goods market in the period after the disaster. For the second hypothetical disaster scenario, we assumed that the government pays the total amount of the damages incurred by the households as disaster aid since this is a standard practice in many developed countries. The availability of capital goods in the market is the only constraint for households.

5.2. Additional Input Data for the Disaster Case

For a basic simulation of a post-disaster economy, two additional input data are specifically needed: the sector-wise capital costs incurred by the firms and households and the corresponding number of affected workers. With these two inputs, the capital losses can be assigned to the firm and households agents at the end of the first period of the simulation.

The data on the economic losses brought by the GEJE are available from various sources. However, most of the data available provides an aggregated value of the losses and the number of affected people. The following are the key points of the input data for the disaster scenarios:

1. The data on the direct economic impact of GEJE were compiled from the data available from the Cabinet Office of Japan (CAO), the Ministry of Agriculture, Forestry, and Fisheries (MAFF), and the National Police Agency of Japan (NPAJ). The MAFF provides direct capital damages to the crop cultivation, agricultural services, forestry, and fishery sectors. The CAO provides aggregated direct capital damages to buildings, lifelines utilities, and infrastructure sectors.
2. The capital damages to buildings comprise of capital damages to households due to the loss of their houses and capital damages to firms due to damages to their buildings. The capital damages to firms due to building losses were distributed to all sectors except for the sectors where the capital losses are already directly available as provided by the MAFF and CAO.
3. The data on the number of affected workers for each sector is available from e-Stat. In this data, the sector classification used is the Japan Standard Industrial Classification. To maintain consistency with the other inputs in the model, this sector classification was mapped into e-Stat's IO Table sector classification and the number of affected workers was reassigned to the corresponding equivalent sectors accordingly.

5.3. Simulations of Post-Disaster Economies

The simulation results for the normal case and the two hypothetical disaster cases are presented in Figure 5. The plot on the real production shows that the impact of the disaster at the national level appears small. This is reasonable given that the GEJE, on which the hypothetical disaster was based,

mainly affected municipalities along the Tohoku Pacific coast, whose production comprised only 2.5% of the nation's GDP. These plots also show that the country was able to recover to its normal production level within one financial quarter. Considering that only a smaller portion of the country was affected by the disaster, it is sensible that the production levels recovered quickly when viewing the results at the national scale. The economic shock induced by the disaster is more visible on the plots on the real capital of firms and real consumption by households, where there is a more visible reduction in the firms' capital and household consumption at the end of the first period.

The production levels after the country has recovered from the disaster seem to surpass the levels in the normal scenario. This happens because even if the disaster caused capital losses, the agents immediately prioritize recovering what they lost, inducing an increased demand in the market. With this increased demand, the firms produce more. The firms' production is decided by the demand and the capital, labor, and materials available. However, the current model does not consider the constraints in the real world such as the limitations of skilled labor forces available for recovery, disruptions to transportation due to disaster-induced damages, and capacity limitations of transportation networks. Further, the assumption that the government provides financial support for the recovery of damaged houses generates a sudden spike in demand in the market. However, in the real world, such a sudden spike in demand from households is not possible since government support for repairing houses is very limited, and financial supports such as insurance payments and bank loans require time. Those are some of the possible reasons why the production in the disaster scenarios appears to be larger than in the normal scenario. A similar trend is observed in real consumption by households; with increased production, employment increases, driving up the income of workers. Since the workers spend a portion of their income on consumption, the households' consumption becomes higher.

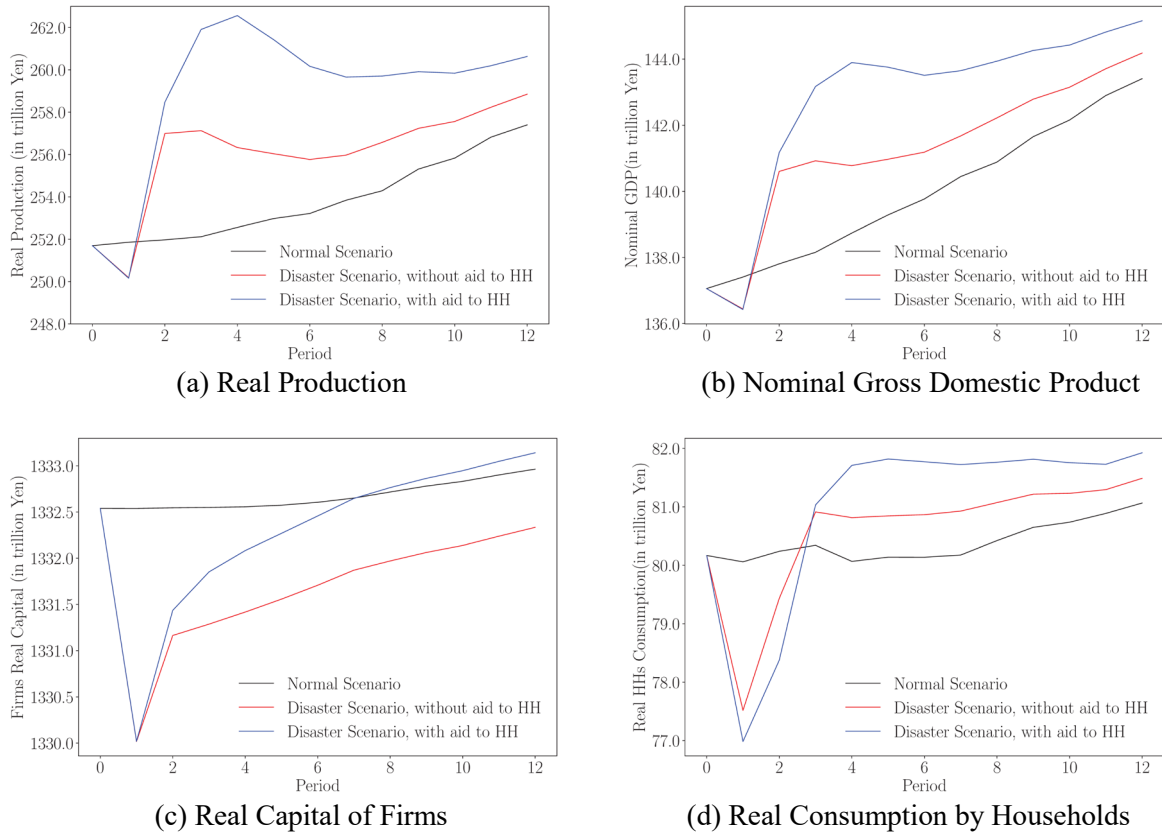


Figure 5. Simulation results on hypothetical post-disaster economies

Upon comparing the two hypothetical disaster scenarios, it can be observed from the results that when the government provides financial aid to households, it leads to a slight increase in household consumption from the 4th period. This outcome is sensible as households have more disposable income when the government provides aid. As a result, households increase their demand for goods from firms,

leading to higher production levels. Since the firms had increased their production, they were also able to increase their capital. In the disaster scenario where the government does not provide financial aid to households, the firms' real capital remains below normal levels, whereas in the scenario where the government provides financial aid to households, firms are able to restore their capital to normal levels by the 7th period.

6. CONCLUDING REMARKS

Comprehensive and reliable estimation of the economic impacts of disasters requires fine-grained modeling of the economy that considers the complex interactions of economic entities with each other and their heavy dependencies on other real-world constraints. Our research attempts to progress the field of disaster risk analysis by developing a large-scale agent-based economic simulator that is capable of simulating economies with hundreds of millions of agents at a 1:1 scale. This paper outlines the development of the tool by utilizing HPC strategies and presents the data generation done to simulate the Japanese economy using publicly available macroeconomic data from various sources and firm-level data from TDB. The validation tests show that the developed framework can replicate past observed macroeconomic data with substantial accuracy at all three levels of resolution: national economy, industrial sectors, and individual firms. A demonstrative application of the developed system shows its potential in closely monitoring various economic indicators of the post-disaster economy in the short to medium term. The model can be further improved by integrating it with other infrastructure components such as lifeline accesses and transportation networks and by using more realistic post-disaster behavioral rules of the economic entities. The developed system is a useful tool in identifying critical portions of the economy, which can be helpful in making more effective recovery plans and in accelerating economic recovery from disasters.

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REFERENCES

- [1] Technical Review Committee on Ensuring Resilience, Japan Society of Civil Engineers (2018): Countermeasures against Huge Disasters that cause National Crises (in Japanese). https://committees.jsce.or.jp/chair/system/files/本編_「国難」をもたらす巨大災害対策についての技術検討報告書_6.pdf (last accessed on July 31, 2023)
- [2] Assenza T., Gatti DD., and Grazzini J. (2015): Emergent dynamics of a macroeconomic agent-based model with capital and credit. *Journal of Economic Dynamics and Control*, 50, 5-28
- [3] Gatti DD., Gallegati M., Greenwald B., Russo A., and Stiglitz J. (2010): The financial accelerator in an evolving credit network. *Journal of Economic Dynamics and Control*, 69, 375-408
- [4] Poledna S., Hochrainer-Stigler S., Miess MG., Klimek P., Schmelzer S., Sorger J., Shchekinova E., Rovenskaya E., Linnerooth-Bayer J., Dieckmann U., Thurner S. (2018): When does a disaster become a systematic event? Estimating indirect economic losses from natural disasters. arXiv:1801.09740
- [5] Poledna S., Miess Michael G., Hommes Cars H. (2019): Economic Forecasting with an Agent-Based Model. <http://dx.doi.org/10.2139/ssrn.3484768>
- [6] Deissenberg C., van der Hoog S., Dawid H. (2008): EURACE: A massively parallel agent-based model of the European economy. *Applied Mathematics and Computation*, 204 541-552
- [7] Gill A., Madgededara L., Poledna S., Hori M., Fujita K., Ichimura T. (2021): High-Performance Computing Implementations of Agent-Based Economic Models for Realizing 1:1 Scale Simulations of Large Economies. *IEEE Transactions on Parallel and Distributed Systems*, 32, 8 (2021), 2101–2114.
- [8] Ministry of Internal Affairs and Communications of Japan (2021): 2015 Input-Output Tables for Japan. https://www.soumu.go.jp/main_content/000732915.pdf (last accessed on July 31, 2023)